Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai : 400058, India

(An Empowered Autonomous College Affiliated to University of Mumbai)

End Semester Examination

December 2024

Max. Marks: 100 | Duration: 3 hours

Class: Third Year | Semester: V

Course Code: CS303 | Program: CE

Name of the Course: Artificial Intelligence and Machine Learning

Instructions:

(1) All questions are compulsory

(2) Draw neat diagrams and write equations wherever necessary

(3) Assume suitable data if necessary

Course Outcomes (CO): At the End of the course students will be able to

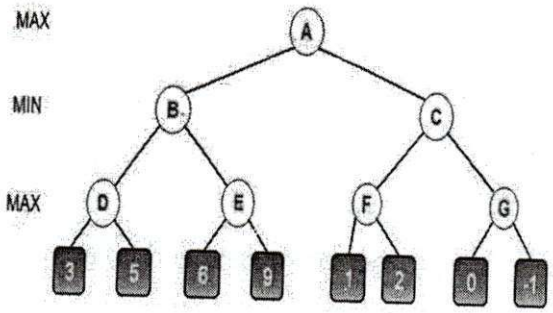
CO1 Understand AI building blocks presented in intelligent agents

CO2 Solve the problems using suitable searching methods.

CO3 Solve the problems using suitable reasoning and knowledge representation methods.

CO4 Apply suitable machine learning technique for a given problem

CO5 Design an intelligent system using different AIML techniques for real life problems.

Q. No		Max. Marks	CO-BL-PI
Q.1a	Is AI a science, or is it engineering? Or neither or both? Explain.	10	1-3-1.3.1
Q.1b	What are the properties of Task Environment ? Give examples of each.	10	1-2-1.3.1
Q.2a	Enumerate Classical "Water jug Problem". Describe the state space for this problem and give the solution.	10	2-3-1.4.1
Q.2b	Apply alpha beta pruning algorithm to figure given below. 	10	2-3-2.1.2

<table>
<tr><td>Q.3a</td><td>Translate the following sentences to propositional logic:
i) I like cake but I don't like cupcakes.
ii) On Saturday, Jane goes for a run or plays basketball, but not both.
iii) If it is hot today, then I'll get ice cream.
iv) You can login to a CS lab computer if you have a CS account.
v) Wilma eats cookies only if Evelyn makes them.</td><td>10</td><td>3-3-1.1.1</td></tr>
<tr><td>Q.3b</td><td>Explain unification algorithm used for reasoning under predicate logic with an example.</td><td>10</td><td>3-3-2.1.3</td></tr>
<tr><td>Q.4a</td><td>Cluster the following eight points (with (x, y) representing locations) into three clusters A1(2, 10) A2(2, 5) A3(8, 4) A4(5, 8) A5(7, 5) A6(6, 4) A7(1, 2) A8(4, 9). Initial cluster centers are : A1(2, 10), A4(5, 8) and A7(1, 2). Use k-means algorithm to find the three cluster centers after the second iteration.</td><td>10</td><td>4-3-2.1.2</td></tr>
<tr><td>Q.4b</td><td>Using a given training set. Create classification model using decision tree classify <income=medium, credit=good>
<table><tr><th>T-id</th><th>Income</th><th>Credit</th><th>Decision</th></tr><tr><td>1</td><td>Very High</td><td>Excellent</td><td>Authorize</td></tr><tr><td>2</td><td>High</td><td>Good</td><td>Authorize</td></tr><tr><td>3</td><td>Medium</td><td>Excellent</td><td>Authorize</td></tr><tr><td>4</td><td>High</td><td>Good</td><td>Authorize</td></tr><tr><td>5</td><td>Very High</td><td>Good</td><td>Authorize</td></tr><tr><td>6</td><td>Medium</td><td>Excellent</td><td>Authorize</td></tr><tr><td>7</td><td>High</td><td>Bad</td><td>Request ID</td></tr><tr><td>8</td><td>Medium</td><td>Bad</td><td>Request ID</td></tr><tr><td>9</td><td>High</td><td>Bad</td><td>Reject</td></tr><tr><td>10</td><td>Low</td><td>Bad</td><td>Call Police</td></tr></table></td><td>10</td><td>4-3-2.1.2</td></tr>
<tr><td>Q.5a</td><td>What is K Nearest neighbor algorithm? How do you choose an optimal K in K-NN? Why KNN is lazy learner?</td><td>10</td><td>4-4-2.4.3</td></tr>
<tr><td>Q.5b</td><td>What are outliers? How do you detect and treat them? How do you deal with outliers in a linear regression model?
OR
What are shrinkage methods, and how are they used in machine learning? Why is Shrinkage Important?</td><td>10</td><td>4-4-2.4.3</td></tr>
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Course outcome	Marks in ESE	Marks in MSE	Total marks for each course outcome
CO1	20	8	28
CO2	20	16	36
CO3	20	6	26
CO4	40		40
CO5	-	-	-